Robot Maintenance Analysis - Submission

Student Details:

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Note: Roshan was unable to attend due to illness.

Group: 2

Project Overview:

This project analyzes Robot Maintenance Data, specifically focusing on the current consumption of different robot axes. By applying Data Engineering pipelining and Linear Regression, we aim to identify correlations between axes (e.g., Axis #1 and Axis #6), which is critical for anomaly detection and predictive maintenance.

System Architecture & Structure:

- data/raw/: robot_maintenance_data.csv (RMBR4-2 raw export).
- data/processed/: robot_current_clean.csv (Cleaned CSV) & robot_data.db (SQLite).
- notebooks/EDA.ipynb: Data ingestion, DB storage, and Time Series Analysis.
- notebooks/linear_regression.ipynb: Modeling axis relationships using custom gradient descent.
- src/: Reusable modules.
- experiments/: Results logging.

Core Implementation Details:

1. Data Pipeline: We ingest raw CSV data, filter for 'current' traits, parse timestamps, and handle missing values.
2. EDA: We visualized Axis #1 current over time and calculated a correlation matrix across all 14 axes.
3. custom Linear Regression: We implemented a Scratch Linear Regression model to predict Axis #6 current based on Axis #1. This relationship helps model normal operating behavior.
4. Validation: We benchmarked our model against Scikit-Learn, achieving comparable RMSE.

Execution Instructions:

1. Install dependencies: pip install -r requirements.txt
2. Run EDA: Execute notebooks/EDA.ipynb to process the robot data.
3. Run Modeling: Execute notebooks/linear_regression.ipynb.
4. View Results: Check experiments/results.csv.

Repository:

https://github.com/alicih4n/DataStreamVisualization_Workshop.git